

Requirement Analysis Phase

Introduction

The **Requirement Analysis Phase** is an important stage in the development of the **EduGenie – Google Gemini Powered Learning Assistant**. In this phase, the requirements of the system are identified and analyzed based on the needs of students. It helps define what the application should do and what resources are required to develop it.

1. Functional Requirements

The functional requirements describe the main activities that the EduGenie system should perform:

- Students should be able to enter their questions or doubts.
- The system should process the questions using Google Gemini.
- It should generate relevant and understandable answers.
- The system should provide step-by-step explanations when required.
- Students should be able to ask follow-up questions.
- The application should support different academic subjects.
- The system should display responses in a clear and readable format.
- It should provide quick responses to student queries.

2. Non-Functional Requirements

The system should also satisfy certain quality requirements:

- **Easy to Use:** The interface should be simple and user-friendly.
- **Fast Response:** Answers should be generated within a reasonable time.
- **Accuracy:** The system should provide relevant and reliable information.
- **Security:** User information and data should be handled safely.
- **Reliability:** The application should work consistently.
- **Scalability:** The system should be capable of supporting more users in the future.

3. Hardware Requirements

The basic hardware requirements include:

- Computer or laptop
- Minimum 4 GB RAM
- Internet connection

- Keyboard and mouse
- Smartphone or other device for accessing the application, if required

4. Software Requirements

The software requirements may include:

- Operating System: Windows/Linux/macOS
- Programming language and suitable development framework
- Web browser such as Chrome or Edge
- Internet connection
- Google Gemini API
- Database, if user data or conversation history needs to be stored

5. User Requirements

The primary users of EduGenie are **students**. They should be able to interact with the application without requiring advanced technical knowledge. Students should be able to enter questions, read AI-generated explanations, and continue the conversation to clarify their doubts.

6. Expected Result

The requirement analysis provides a clear understanding of the features, resources, and technical needs of the project. It acts as a foundation for the next stages of development and helps ensure that **EduGenie meets the educational needs of students effectively**.